

Integer sums of two units in the seventh cyclotomic field

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Abstract

Let $K = \mathbb{Q}(\zeta_7)$. The positive rational integers that are sums of two units of $\mathcal{O}_K$ are exactly 1, 2, 3, 4, 5, 19, 22. Moreover, every representing pair with sum greater than one consists of units in the maximal real subfield. We prove this realness assertion by the standard cyclotomic unit decomposition and a two-equation norm calculation, then apply the established cyclic-cubic classification of Tinková, Visser and Yatsyna. Explicit unit pairs and integral inverses certify all seven values. The real-cubic classification is an external input; no new classification of real-cubic units or priority claim is made.

1 Statement and the known cubic input

For a number field L , write $\mathcal{O}_L$ for its full ring of integers and set

$$N_L = \{n \in \mathbb{Z}_{>0} : n = \varepsilon + \delta \text{ for some } \varepsilon, \delta \in \mathcal{O}_L^\times\}.$$

The two units may coincide, and neither is required to be positive under any real embedding. Throughout, put

$$z = \exp(2\pi i/7), \quad K = \mathbb{Q}(z), \quad t = z + z^{-1}, \quad F = \mathbb{Q}(t).$$

Thus $\mathcal{O}_K = \mathbb{Z}[z]$ and F is the maximal real subfield of K . The prime-cyclotomic question posed by Newman is reviewed by Visser and Ziegler [3]. We give the following explicit seventh-cyclotomic deduction.

Theorem 1. *One has*

$$N_K = N_F = \{1, 2, 3, 4, 5, 19, 22\}. \tag{1}$$

If $\varepsilon, \delta \in \mathcal{O}_K^\times$ and $\varepsilon + \delta = n \in \mathbb{Z}_{>1}$, then $\varepsilon, \delta \in \mathcal{O}_F^\times$. For each integer in (1), Table 1 supplies a representing pair and integral inverses.

The classification over F is already known. In the notation of Tinková–Visser–Yatsyna [2, Theorem 1.2], the field K_a is generated by a root of

$$f_a(X) = X^3 - aX^2 - (a+3)X - 1.$$

Their theorem gives

$$N_{K_{-1}} = \{1, 2, 3, 4, 5, 19, 22\}. \tag{2}$$

Indeed, $t^3 + t^2 - 2t - 1 = 0$, and this polynomial has no rational root, so $F = K_{-1}$. The field is cyclic of degree three, being the fixed field of conjugation in the cyclic extension $K/\mathbb{Q}$. Thus all the units of the full ring $\mathcal{O}_F$ are covered by (2).

The step needed to transfer this list to K is the exclusion of every additional nonreal representation when $n > 1$. The unit decomposition used below is standard; it is recalled, for example, by Ailon and Rudnick [1, Section 4, equation (3)]. We include its proof in this field, together with the norm exclusion, so that the only substantial external classification input is (2).

2 Unit descent and exclusion of nonreal pairs

Lemma 2. *Every $\varepsilon \in \mathcal{O}_K^\times$ can be written uniquely as $\varepsilon = z^a u$, where $a \in \mathbb{Z}/7\mathbb{Z}$ and $u \in \mathcal{O}_F^\times$.*

Proof. First, the roots of unity in K are exactly $\{\pm z^j : 0 \leq j < 7\}$. To verify this, $\text{Gal}(K/\mathbb{Q})$ is cyclic of order six and has just one subgroup of index two. The element

$$g = z + z^2 + z^4 - z^3 - z^5 - z^6$$

satisfies $g^2 = -7$, by reduction modulo $1 + z + \cdots + z^6 = 0$. Hence the unique quadratic subfield is $\mathbb{Q}(\sqrt{-7})$. If a primitive m th root lies in K , then $\varphi(m)$ divides six. The possibilities are

$$m \in \{1, 2, 3, 4, 6, 7, 9, 14, 18\}.$$

For completeness, each prime $q \mid m$ has $q - 1 \mid 6$, so only 2, 3, 7 can occur; the condition on $\varphi(m)$ bounds their exponents and gives the displayed list. Orders divisible by three would give $\mathbb{Q}(\sqrt{-3}) \subset K$, and order four would give $\mathbb{Q}(i) \subset K$. The unique quadratic subfield excludes these possibilities. Only orders 1, 2, 7, 14 remain.

Now let $\alpha = \varepsilon/\bar{\varepsilon}$. This is an algebraic integer, since ε and $\bar{\varepsilon}^{-1}$ are integral. Every embedding $\sigma : K \hookrightarrow \mathbb{C}$ commutes with conjugation, so $|\sigma(\alpha)| = 1$. Here is the needed form of Kronecker's argument. For every positive integer k , the polynomial

$$P_k(X) = \prod_{\sigma: K \hookrightarrow \mathbb{C}} (X - \sigma(\alpha)^k)$$

has integer coefficients: they are algebraic integers fixed by the Galois group. The absolute values of its coefficients are bounded by the binomial coefficients of degree six. There are therefore only finitely many possible polynomials P_k , and their roots form a finite set. Consequently two positive powers of α agree, and α is a root of unity.

The quotient $\mathcal{O}_K/(1-z)$ is $\mathbb{F}_7$, since substituting $z = 1$ in $\Phi_7(z) = 0$ gives $\Phi_7(1) = 7$. Conjugation acts identically on this quotient. A unit has nonzero residue, so α reduces to 1. The roots $-z^j$ reduce to $-1 \neq 1$, and hence $\alpha = z^k$ for some $k \in \mathbb{Z}/7\mathbb{Z}$. Choose a with $2a = k$ modulo seven and set $u = z^{-a}\varepsilon$. Then

$$\bar{u} = z^a \bar{\varepsilon} = z^{a-k} \varepsilon = z^{-a} \varepsilon = u.$$

Thus $u \in F$. Both u and u^{-1} are integral, so both lie in $F \cap \mathcal{O}_K = \mathcal{O}_F$, proving $u \in \mathcal{O}_F^\times$. Conversely, $z^a u$ is a unit whenever $u \in \mathcal{O}_F^\times$. Uniqueness follows since 1 is the only real seventh root of unity. $\square$

Lemma 3. *If $\varepsilon + \delta = n \in \mathbb{Z}_{>0}$ with $\varepsilon, \delta \in \mathcal{O}_K^\times$, then either both units lie in $\mathcal{O}_F^\times$ or $n = 1$.*

Proof. By Lemma 2, write $\varepsilon = z^a u$ and $\delta = z^b v$ with $u, v \in \mathcal{O}_F^\times$ and $0 \leq a, b < 7$. The equation and its conjugate give

$$z^a u + z^b v = n, \quad z^{-a} u + z^{-b} v = n. \quad (3)$$

If $a = b = 0$, both units are real. If exactly one phase is zero, subtracting the equations forces the other real unit to vanish, which is impossible. If $a = b \neq 0$, subtraction gives $u + v = 0$, hence $n = 0$, also impossible. The determinant vanishes precisely when $a = b$, since $z^{a-b} = z^{b-a}$ is equivalent to $2(a - b) = 0$ modulo seven.

It remains to consider distinct nonzero a, b . Solving (3) gives

$$\varepsilon = nc, \quad c = \frac{z^a(z^b - z^{-b})}{z^{b-a} - z^{a-b}}. \quad (4)$$

For $m \not\equiv 0 \pmod{7}$,

$$N_{K/\mathbb{Q}}(1 - z^m) = \prod_{j=1}^6 (1 - z^{mj}) = \Phi_7(1) = 7. \quad (5)$$

Every difference $z^h - z^{-h}$ with $h \not\equiv 0 \pmod{7}$ is a root of unity times $1 - z^{2h}$, up to sign. Since $N_{K/\mathbb{Q}}(z) = N_{K/\mathbb{Q}}(-1) = 1$, the numerator and denominator of c both have norm seven. Hence $N_{K/\mathbb{Q}}(c) = 1$. The norm of a unit in the totally imaginary field K is positive and equals one: its embeddings occur in conjugate pairs, and its norm is an integer unit. Taking norms in (4) now yields

$$1 = N_{K/\mathbb{Q}}(\varepsilon) = n^6 N_{K/\mathbb{Q}}(c) = n^6.$$

The positivity of n forces $n = 1$. No integrality assertion about c is needed to take its field norm. $\square$

3 Witnesses and completion of the classification

Table 1: For each row, put $\delta = n - \varepsilon$ and $t = z + z^{-1}$. The last two columns are integral inverses.

n	ε	ε^{-1}	δ^{-1}
1	t	$t^2 + t - 2$	$-t^2 - 2t$
2	1	1	1
3	t^2	$-2t^2 - t + 5$	$-t^2 + t + 1$
4	$t^2 + 2t$	$t - 1$	$-4t^2 - 9t - 3$
5	$t^2 - t$	$t + 2$	$-5t^2 + 4t + 3$
19	$5t^2 + 9t$	$14t^2 - 11t - 8$	$81t^2 + 182t + 65$
22	$4t^2 - 5t$	$-13t^2 - 29t - 10$	$101t^2 - 81t - 56$

The nontrivial real-field witnesses agree with those in [2, Table 3]. Each row of Table 1 is verified by multiplying the two proposed inverse pairs and reducing by

$$t^3 = -t^2 + 2t + 1. \quad (6)$$

Both products equal 1. All displayed expressions belong to $\mathbb{Z}[t]$, so they are algebraic integers. The identities therefore certify that ε and $n - \varepsilon$ are units of $\mathcal{O}_F$, without requiring an integral-basis calculation.

Proof of Theorem 1. The table gives the inclusion of all seven integers in $N_F \subseteq N_K$. Conversely, let $n \in N_K$. If $n > 1$, Lemma 3 places every representing pair in $\mathcal{O}_F^\times$, and the known classification (2) places n in the displayed set. If $n = 1$, it is already in that set. This proves both assertions. $\square$

The strict hypothesis $n > 1$ in the realness assertion is necessary. For example,

$$(1 + z^2) + (-z^2) = 1$$

is a nonreal unit pair. Indeed, $1 + z^2 = (1 - z^4)/(1 - z^2)$ has norm one by (5) and is an algebraic integer, hence is a unit. We do not classify all pairs at $n = 1$ or address other prime conductors.

Exact verification and scope. The accompanying Python program `verify_witnesses.py` checks all fourteen inverse identities using integer arithmetic in $\mathbb{Z}[T]/(T^3 + T^2 - 2T - 1)$; it needs no external package. The supplementary checks certify these finite identities, not the global cubic classification or an exponent bound. There is no height cutoff or finite search assumption in the proof. Exact polynomial certificates also occur in Diophantine-tuple work such as [4]; that reference is adjacent methodology and is not an input to the unit-equation theorem.

This note is a short deduction from the established cubic result and the standard unit decomposition, with a complete phase-norm exclusion in K . The consulted primary sources did not supply an explicit statement of this seventh-cyclotomic conclusion. This limited literature comparison does not establish priority or that the question was previously open; in particular, the full older articles of Newman and Kostra were not inspected.

Computational and AI assistance. OpenAI Codex assisted with source comparison, development and checking of the algebraic argument, exact verification programs, manuscript drafting, and a separate model review. Exact integer computations checked the seven witness pairs and the thirty distinct nonzero phase choices; an additional implementation used cyclic convolution and products over the six Galois embeddings. These computations verify finite identities only. The external cyclic-cubic theorem is relied upon rather than independently reproved. The model review is not human peer review or formal proof verification.

References

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